

Comeback Momentum, Chart Re-Entry, and Fandom Intensity Analysis of South Korea Top 50 Spotify Playlist

Internship Research Paper

Submitted by:

Hamshalekha N.J

Tools Used:

- Python
- Google Colab
- Pandas
- NumPy
- Plotly Express
- Streamlit

Abstract

The South Korea music industry, particularly K-Pop, is characterized by frequent artist comebacks, highly dedicated fandoms, and intense streaming activities. Unlike many global music markets where chart performance depends mainly on long-term popularity, the Korean music ecosystem experiences repeated momentum spikes caused by album releases, promotional campaigns, concerts, anniversaries, and fandom-driven streaming events. This project investigates comeback momentum, chart re-entry behavior, and fandom intensity using the Spotify South Korea Top 50 Playlist dataset.

The analysis includes data preprocessing, exploratory data analysis, feature engineering, momentum score calculation, and visualization techniques to understand how artists maintain chart success over time. A custom Momentum Score was developed using popularity, chart retention, and re-entry frequency. The results reveal that sustained chart performance is strongly influenced by fan engagement and promotional strategies rather than popularity alone. An interactive Streamlit dashboard was developed to allow users to explore artist performance, momentum distribution, album types, and fandom trends.

Introduction

The rapid growth of the Korean music industry has transformed K-Pop into a global entertainment phenomenon. Artists from South Korea consistently dominate music charts across multiple streaming platforms due to strong fan communities and well-planned promotional activities.

Unlike traditional music industries where chart rankings change gradually, K-Pop songs frequently experience sudden increases in popularity after comeback events, music shows, live performances, social media trends, or fan-organized streaming campaigns. These events create momentum spikes that significantly influence chart rankings.

This project aims to analyze these unique patterns by studying Spotify's South Korea Top 50 Playlist. Instead of focusing solely on popularity scores, the project introduces a Momentum Score that combines multiple performance indicators, providing a more comprehensive understanding of song success and fandom influence.

Background

South Korea's music ecosystem is unique because fan participation plays a major role in determining chart success. Fans actively organize streaming events, purchase albums in large quantities, and promote artists across social media platforms. As a result, songs often return to music charts multiple times after their initial release.

Several important characteristics distinguish the South Korean music market:

- Strong fandom-driven streaming behavior
- Frequent artist comeback promotions
- Multiple chart re-entries
- Album release cycles generating momentum spikes
- High engagement during concerts, anniversaries, and award shows

These characteristics make momentum analysis more meaningful than analyzing popularity scores alone.

Problem Statement

Although Spotify provides popularity scores for songs, popularity alone does not fully explain why certain songs remain successful or repeatedly return to music charts. Traditional ranking metrics fail to capture the influence of comeback events, fan-driven streaming campaigns, and long-term audience engagement.

Therefore, this project aims to develop a momentum-based analytical framework that measures chart performance using multiple indicators instead of relying solely on popularity.

Objectives

The primary objectives of this project are:

- Analyze the South Korea Top 50 Spotify Playlist.
- Measure comeback momentum for individual songs.
- Identify songs that repeatedly re-enter music charts.
- Compare artist performance using momentum scores.
- Evaluate fandom engagement through chart behavior.
- Study album type distributions.
- Analyze explicit versus non-explicit songs.
- Develop an interactive Streamlit dashboard for visualization.
- Generate business insights for music companies and record labels.

Dataset Description

The dataset used in this project contains Spotify South Korea Top 50 Playlist records collected over multiple dates. Each row represents the appearance of a song in the playlist.

Column	Description
song	Song title
artist	Artist name
avg_popularity	Average Spotify popularity score
total_days	Total number of days the song remained on the chart
reentry_count	Number of times the song re-entered the playlist
momentum_score	Calculated momentum score
momentum_score_final	Normalized momentum score
album_type	Album or Single
is_explicit	Indicates whether the song contains explicit content
Momentum_Level	Categorized momentum level (Low, Medium, High)

The dataset was cleaned and transformed before performing statistical analysis and visualization.

Methodology

This project follows a structured data analytics workflow consisting of data collection, preprocessing, feature engineering, exploratory data analysis, visualization, and dashboard development. The methodology was designed to identify songs with strong comeback momentum and measure fandom engagement through Spotify chart performance.

The major stages of the methodology are:

Step 1: Data Collection

The Spotify South Korea Top 50 Playlist dataset was collected and imported into Google Colab for analysis. The dataset contains daily chart records, including song title, artist name, popularity score, album type, explicit content, and ranking information.

Step 2: Data Cleaning

The raw dataset was examined for missing values, duplicate records, and inconsistent data types. Necessary cleaning operations were performed to improve data quality before analysis.

Step 3: Feature Engineering

Several new analytical features were created from the original dataset, including:

- Average Popularity
- Total Days on Chart
- Re-entry Count
- Momentum Score
- Normalized Momentum Score
- Momentum Level (Low, Medium, High)

Step 4: Exploratory Data Analysis

Different statistical and graphical techniques were applied to understand relationships among popularity, momentum, album type, and artist performance.

Step 5: Dashboard Development

A Streamlit web application was developed to provide an interactive dashboard where users can explore songs, artists, momentum levels, and album distributions.

Data Cleaning

Data cleaning is one of the most important stages of any data analytics project because inaccurate or inconsistent data can produce misleading results.

The following preprocessing techniques were performed:

7.1 Removing Missing Values

Missing values were identified using Pandas functions and removed whenever necessary.

```
df.isnull().sum()
```

Rows containing invalid information were excluded from the analysis to improve data quality.

7.2 Removing Duplicate Records

Duplicate rows were identified and removed using:

```
df.drop_duplicates()
```

This ensured that each observation represented a unique chart record.

7.3 Data Type Conversion

The **date** column was converted into DateTime format for chronological analysis.

```
df['date'] = pd.to_datetime(df['date'])
```

This conversion enabled proper sorting and time-based calculations.

7.4 Data Standardization

Text values such as artist names and song titles were standardized to maintain consistency throughout the dataset.

Feature Engineering

Feature engineering was performed to extract meaningful information from the raw Spotify dataset.

The following new variables were created.

Average Popularity

Average popularity was calculated for every song across all available chart entries.

This metric represents the long-term audience interest rather than a single-day popularity score.

Total Days on Chart

The total number of days each song remained in the Spotify Top 50 playlist was calculated.

Songs appearing for longer durations indicate sustained audience engagement.

Re-entry Count

Some songs disappeared from the chart and later returned due to promotional activities, fan streaming events, or comeback performances.

The total number of re-entries was calculated for each song.

Higher re-entry values indicate stronger comeback potential.

Momentum Score

A custom Momentum Score was developed by combining multiple performance indicators.

The score considers:

- Average Popularity
- Total Days on Chart
- Re-entry Count

This composite score provides a better representation of overall song performance than popularity alone.

Momentum Score Normalization

Since different variables have different scales, normalization was applied.

The normalized score ranges between **0 and 1**, making comparison across songs easier.

This normalized value is stored as:

momentum_score_final

Momentum Level

Songs were categorized into three momentum groups.

Momentum Score	Category
Less than 0.40	Low
0.40 – 0.74	Medium
0.75 and above	High

This classification simplifies interpretation and visualization.

Momentum Score Formula

The Momentum Score is the key metric developed in this project.

It combines multiple indicators representing popularity, chart longevity, and comeback behaviour.

The formula is expressed as:

Momentum Score=(Average Popularity)+(Total Days on Chart)+(Re-entry Count)
$$\text{Momentum Score} = (\text{Average Popularity}) + (\text{Total Days on Chart}) + (\text{Re-entry Count})$$

The calculated score was subsequently normalized using Min-Max Scaling to generate the final Momentum Score.

Normalization Formula:

Normalized Score=
$$\frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

The normalized value enables direct comparison between songs regardless of differences in scale.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns, trends, and relationships in the dataset.

Several statistical summaries and visualizations were generated.

Top Artists by Momentum Score

Artists were ranked according to the average Momentum Score of their songs.

This analysis identified artists with the strongest chart performance and most consistent audience engagement.

Momentum Distribution Analysis

Histogram analysis showed how Momentum Scores are distributed among songs.

The distribution revealed whether most songs possess low, medium, or high momentum.

Box plots were also used to identify potential outliers.

Momentum Level Analysis

Songs were classified into Low, Medium, and High Momentum categories.

Pie charts and bar charts illustrated the proportion of songs in each category.

This analysis provides a simplified overview of overall chart performance.

Rank Recovery Analysis

Rank Recovery measures the ability of songs to regain higher chart positions after experiencing ranking declines.

Songs with higher recovery values demonstrate stronger resilience and sustained audience support.

Album Type Analysis

The dataset was analyzed based on album categories.

Album Type Distribution identifies whether Singles or Albums dominate the South Korea Top 50 Playlist.

Explicit Content Analysis

The percentage of Explicit and Non-Explicit songs was calculated.

This provides insights into listener preferences and content trends.

Top Songs Analysis

The Top 10 songs with the highest Momentum Scores were identified.

These songs represent the strongest performers in terms of popularity, longevity, and comeback momentum.

Visualizations

Several interactive visualizations were developed using Plotly Express.

The major visualizations include:

- Dataset Overview Table
- Top Artists by Momentum Score
- Momentum Score Distribution
- Momentum Level Distribution
- Album Type Distribution
- Explicit vs Non-Explicit Songs
- Top 10 Momentum Songs
- KPI Dashboard
- Interactive Streamlit Dashboard

These visualizations improve understanding of chart behaviour and help identify important performance trends quickly.

Results and Discussion

The analysis of the South Korea Top 50 Spotify Playlist provided valuable insights into song performance, comeback momentum, and fandom engagement. By combining popularity, chart longevity, and re-entry frequency into a unified Momentum Score, the project successfully identified songs and artists that maintained strong chart performance over time.

The Top Artists by Momentum Score analysis revealed that certain artists consistently achieved higher momentum due to strong fan support and successful promotional activities. Rather than depending solely on popularity, these artists demonstrated sustained chart presence through repeated engagement with listeners.

The Momentum Distribution Analysis indicated that most songs fell into the Medium Momentum category, while a smaller number achieved High Momentum. This suggests that only a limited number of songs sustain exceptional performance over long periods, whereas the majority maintain moderate chart success.

Album Type Analysis showed noticeable differences between Singles and Albums. Singles generally dominated the playlist because they are often released as promotional tracks and receive higher streaming activity during comeback periods.

Explicit Content Analysis indicated that non-explicit songs represented a larger portion of the playlist, reflecting the listening preferences of a broad audience in South Korea.

The Rank Recovery Analysis demonstrated that several songs regained stronger chart positions after temporary declines, highlighting the influence of fan-organized streaming events, live performances, and promotional campaigns.

Overall, the Momentum Score proved to be an effective metric for evaluating long-term chart performance beyond traditional popularity measures.

Business Insights

The findings of this project provide several valuable insights for music streaming platforms, entertainment companies, artists, and marketing teams.

Entertainment Companies

Entertainment agencies can use Momentum Scores to evaluate the effectiveness of comeback strategies. Artists with consistently high momentum may require fewer promotional campaigns, whereas artists with declining momentum may benefit from additional marketing efforts.

Music Streaming Platforms

Streaming services can recommend songs with strong momentum to users, increasing listener engagement and playlist quality. Momentum-based recommendation systems may provide more accurate suggestions than popularity-based systems alone.

Marketing Teams

Marketing professionals can identify the optimal timing for promotional campaigns by monitoring momentum trends and chart recovery behavior.

Artist Management

Artist managers can use chart longevity and momentum analytics to evaluate fan engagement and measure the success of promotional activities.

Record Labels

Record labels can identify high-performing artists and allocate promotional budgets more efficiently based on Momentum Scores instead of relying solely on streaming counts.

Limitations

Although the project successfully analyzed chart momentum, several limitations should be considered.

- The analysis was limited to the Spotify South Korea Top 50 Playlist.
- External factors such as YouTube views, TikTok trends, television appearances, and live concerts were not included.
- Listener demographic information was unavailable.
- Sentiment analysis of fan comments and social media posts was not performed.
- The Momentum Score was developed specifically for this project and may require further validation using larger datasets.

Despite these limitations, the project provides meaningful insights into music chart behavior and fan engagement.

Future Scope

Several improvements can be incorporated into future versions of this project.

- Integrate Spotify API for real-time data collection.
- Include YouTube Music, Apple Music, and Melon chart data.
- Apply Machine Learning algorithms to predict future chart rankings.
- Perform sentiment analysis using Twitter, Reddit, and YouTube comments.
- Develop advanced recommendation systems based on momentum patterns.
- Analyze seasonal trends and comeback schedules.
- Build a cloud-hosted dashboard with automatic updates.
- Compare momentum across multiple countries to study global music trends.

These improvements would increase the analytical capabilities of the system and provide more comprehensive insights.

16. Conclusion

This project successfully analyzed the South Korea Top 50 Spotify Playlist by introducing a Momentum Score that combines popularity, chart longevity, and re-entry behavior into a single performance indicator.

The analysis demonstrated that popularity alone does not fully explain long-term chart success. Songs with strong fandom engagement often maintain high momentum despite fluctuations in popularity rankings. The results also highlighted the importance of promotional campaigns, comeback events, and dedicated fan communities in sustaining chart performance.

The developed Streamlit dashboard provides an interactive platform for exploring artist performance, song momentum, album distributions, and fandom intensity. The dashboard enables users to filter data dynamically and visualize important trends through interactive charts.

Overall, this project demonstrates how data analytics can support strategic decision-making in the music industry by identifying successful artists, evaluating promotional effectiveness, and understanding listener behavior. The proposed methodology can also be extended to other music markets and streaming platforms.

17. References

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Appendix

Software Used

- Google Colab
- Python 3
- Pandas
- NumPy
- Plotly Express
- Streamlit
- Microsoft Word

Dataset Used

Spotify South Korea Top 50 Playlist Dataset

Output Generated

- Data Cleaning Report
- Exploratory Data Analysis
- Momentum Score Calculation
- Momentum Distribution Analysis
- Rank Recovery Analysis
- Album Type Analysis
- Explicit Content Analysis
- Top Artists Analysis
- Top Songs Analysis
- KPI Dashboard
- Interactive Streamlit Dashboard

Executive Summary

Executive Summary

Project Title

Comeback Momentum, Chart Re-Entry, and Fandom Intensity Analysis of South Korea Top 50 Spotify Playlist

Overview

The South Korea music industry is one of the fastest-growing entertainment industries in the world, with K-Pop artists achieving global recognition through highly dedicated fan communities and continuous promotional activities. Unlike traditional music charts, songs in the South Korean music market often experience repeated popularity spikes due to comeback events, fan-organized streaming campaigns, concerts, and social media trends.

This project focuses on analyzing Spotify's South Korea Top 50 Playlist to understand chart performance, comeback momentum, and fandom engagement using data analytics techniques.

Objective

The primary objective of this project is to develop a Momentum Score capable of measuring long-term song performance by combining multiple indicators rather than relying solely on Spotify popularity scores.

The project aims to:

- Analyze chart performance
- Measure comeback momentum
- Detect chart re-entry patterns
- Compare artist performance
- Analyze fandom intensity
- Build an interactive dashboard

Dataset

The dataset consists of Spotify South Korea Top 50 Playlist records containing information about:

- Song Name
- Artist
- Popularity Score
- Album Type
- Explicit Content
- Chart History

Additional analytical features were generated during feature engineering.

Methodology

The project followed a systematic data analytics workflow.

The major stages include:

- Data Collection
- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis
- Momentum Score Calculation
- Data Visualization
- Streamlit Dashboard Development

Tools Used

The following tools were used throughout the project:

- Python
- Google Colab
- Pandas
- NumPy
- Plotly Express
- Streamlit
- Microsoft Word

Major Features Developed

The project successfully developed several analytical features including:

- Average Popularity
- Total Days on Chart
- Re-entry Count
- Momentum Score
- Normalized Momentum Score
- Momentum Level Classification

Dashboard Features

A Streamlit dashboard was developed containing:

- KPI Dashboard
- Dataset Preview
- Artist Filter
- Album Filter
- Momentum Level Filter
- Top Artist Analysis
- Momentum Distribution
- Album Distribution
- Explicit Content Analysis
- Top Songs Analysis
- Search Functionality
- Download Filtered Dataset

Key Findings

The analysis produced several important findings.

- Songs with high Momentum Scores generally remained longer on the charts.
- Re-entry frequency is an important indicator of fan engagement.
- Popularity alone does not accurately represent long-term performance.
- Singles dominate playlist rankings compared to albums.
- Most songs fall into the Medium Momentum category.
- Artists with strong fandom communities consistently achieve higher momentum.

Business Impact

This analytical framework can assist:

- Entertainment Companies
- Artist Management Teams
- Music Streaming Platforms
- Marketing Agencies
- Record Labels

in making data-driven decisions regarding promotional campaigns, artist management, and recommendation systems.

Conclusion

The project demonstrates that combining popularity, chart longevity, and re-entry frequency into a Momentum Score provides a more comprehensive evaluation of song performance than traditional popularity metrics. The developed Streamlit dashboard offers an interactive platform for exploring music analytics and supports better strategic decision-making within the music industry.